

LCD L40048: Allergen Immunotherapy (AIT) with Subcutaneous Immunotherapy (SCIT)

Version 6

Coverage Indications, Limitations, and/or Medical Necessity Coverage Guidance

Compliance with the provisions in this LCD may be monitored and addressed through post-payment data analysis and subsequent medical review audits.

History/Background and General Information

Allergen immunotherapy (AIT) involves the administration of an allergen to which the patient is sensitive, for the purpose of modulating the untoward immune response to that allergen and alleviating allergic symptoms. AIT represents the only therapy capable of inducing a state of immune tolerance and, through its inherent disease-modifying properties, provides the potential to affect a sustained clinical benefit with long-lasting clinical remission of the allergic condition. In addition, it offers the possibility of preventing the development of new allergen sensitivities in the allergic patient, inhibiting the progression of allergic rhinitis to asthma, and improving a patient's quality of life and medication requirements. Subcutaneous immunotherapy (SCIT) is the best-established form of this treatment. The indications and efficacy for SCIT with aeroallergens (i.e., inhaled allergens, such as pollens, dust mites, animal dander, etc.) are considered in this LCD. SCIT with other allergens, such as venoms, is not considered in this LCD.

General Guidelines

There are several important considerations that should be addressed to determine if aeroallergen SCIT is appropriate for a specific patient. 1,2 SCIT using preparations of aeroallergens may be indicated in the management of the following disorders: allergic rhinitis and/or allergic conjunctivitis, including seasonal allergic rhinitis and/or conjunctivitis, perennial allergic rhinitis and/or conjunctivitis, and both seasonal and perennial allergic rhinitis and/or conjunctivitis; allergic asthma, including seasonal allergic asthma and perennial allergic asthma; and both allergic rhinitis and/or allergic conjunctivitis and allergic asthma. Clinical studies demonstrate that patients with both asthma and allergic rhinitis derive particular benefit. 3-6

Prior to consideration for AIT, the health care professional should ensure that the patient has maximized environmental control measures and is on an optimal medication regimen. If the patient has not been compliant with medications, the reasons for this should be explored in depth, including documentation to substantiate that the medications are either no longer effective or minimally effective in controlling the symptoms of allergic rhinitis, allergic conjunctivitis, or allergic asthma. SCIT is usually recommended for the treatment of allergic respiratory disease only after a period of pharmacologic management to include intranasal steroids and observation. 7 Medications are relatively easy for most patients to use, and when effective, they provide relief more rapidly than immunotherapy. 7

Per consensus published guidelines from a 2020 rhinitis update, it has been suggested that AIT (subcutaneous or sublingual tablets) be offered through shared decision-making to patients with moderate/severe allergic rhinitis who (1) are not controlled with allergen avoidance and/or pharmacotherapy or (2) choose immunotherapy as the preferred method of treatment (e.g., due to the desire to avoid the adverse effects or long-term use of pharmacotherapy), and/or (3) desire the potential benefit of immunotherapy to prevent or reduce the severity of comorbid conditions, such as asthma. It is suggested that AIT (subcutaneous or sublingual tablets) be considered for patients with controlled mild and moderate asthma with coexisting allergic rhinitis. 8

A period of observation also allows the clinician to monitor the patient's disease over time, which is particularly important for adults with new-onset asthma in whom the differential should include other disorders that may present with respiratory symptoms (e.g., other pulmonary conditions [chronic obstructive lung disease, chronic bronchitis, nonallergic asthma], gastroesophageal reflux disease, and cough caused by chronic rhinosinusitis). These conditions should be considered first before initiating SCIT. If current management is suboptimal, it is reasonable to consider a trial of AIT in patients with significant allergic disease, for any of the following reasons: the severity of the patient's condition, its duration (intermittent versus persistent; seasonal versus perennial), and the impact on work, schooling, and quality of life. These are some of the important factors in deciding whether to initiate a course of SCIT. Tangential to these considerations is an understanding of the patient's anticipated goals which highlights the importance of the establishment of a strong patient-physician relationship and the role of shared decision-making in this process. 6,8,9

Covered Indications

SCIT using preparations of aeroallergens can be considered reasonable and necessary in the management of the following disorders:

Allergic rhinitis and/or allergic conjunctivitis, including:

Seasonal allergic rhinitis and/or conjunctivitis

Perennial allergic rhinitis and/or conjunctivitis

Both seasonal and perennial allergic rhinitis and/or conjunctivitis

Allergic asthma:

Seasonal allergic asthma

Perennial allergic asthma

Both allergic rhinitis and/or allergic conjunctivitis and allergic asthma

Atopic dermatitis (AD) due to dust mites

A patient is a candidate for AIT only if it has been established that there is a clinically important allergic component to their disease. For patients with the disorders listed above, clinical relevance is established by the presence of both of the following:

Symptoms upon natural exposure to the allergen OR inferred when the patient has known exposure to an allergen and a temporal pattern of symptoms that is consistent with occurrence of that allergen, such as rhinitis and conjunctivitis during tree pollen season; AND

The presence of specific immunoglobulin E (IgE) to that allergen, demonstrated either through allergen skin testing or serum tests for allergen specific IgE.

Clinical Indications for Allergen Immunotherapy

These clinical indications must be met to be considered reasonable and necessary for AIT:

Symptoms of allergic rhinitis, allergic conjunctivitis, allergic asthma, or any combination of these disorders after natural exposure to aeroallergens AND

Demonstrable evidence of clinically relevant specific IgE AND

At least 1 of the following:

Poor response to pharmacotherapy, allergen avoidance, or both for a minimum of 28 consecutive days

Unacceptable adverse effects of medications

Avoidance of long-term pharmacotherapy and its side effect(s)

Possible prevention of asthma in patients with allergic rhinitis

OR:

For patients with moderate to severe AD due to dust mites who have failed medical management for a minimum of 90 days.

Dosing

The effectiveness of SCIT is dose dependent. The optimal dose or dose range is specific to each type of allergen and varies significantly among allergens. AIT dosing is not adjusted for patient size or age. Children have customarily been given the same dose as adults. Some patients may not tolerate this dose due to repeated large local or systemic reactions. In this situation, the patient's highest tolerated dose becomes their maintenance dose. Most patients placed on immunotherapy are sensitized to multiple allergens when dosed from a mixture of allergens; however, monotherapy may also be reasonable and necessary. 1,10-13

Maintenance Therapy

Time to onset of benefit - The beneficial effects of SCIT for allergic respiratory disease begin during the first year of therapy and continue throughout the period in which the patient receives injections. 1,14-16

Clinical improvement can be demonstrated very shortly after the patient reaches a maintenance dose. 1,17-20

Duration of therapy - There is consensus that an initial course of immunotherapy should consist of 3 to 5 years of maintenance treatment. After this, the clinician and patient should meet to review overall impact on quality of life and based upon these factors, decide if treatment will be continued. 1,16,21,22

A decision about continuation of effective immunotherapy should generally be made after the initial period of 3 to 5 years of treatment. Some patients might experience sustained clinical remission of their allergic disease after discontinuing immunotherapy, but others might relapse. The severity of disease and benefits sustained from treatment are factors that should be considered in determining whether to continue or stop immunotherapy for any individual patient. Patients should be evaluated at least every 6 to 12 months while receiving immunotherapy to assess efficacy, to implement and reinforce its safe administration and to monitor adverse reactions, to assess the patient's compliance with treatment, to determine whether immunotherapy can be discontinued, and to determine whether adjustments in the immunotherapy dosing schedule or allergen content are necessary. 1

Limitations for Allergen Immunotherapy

The following are considered not reasonable and necessary:

First-line treatment for allergic rhinitis and allergic conjunctivitis in the absence of previous medical treatment and/or environmental avoidance

Absence of clinically relevant IgE

AD not due to dust mites

Sublingual immunotherapy (SLIT)

SCIT during pregnancy

Treatment for food sensitivities

A presumption of failure can be made when, after 12-24 months of therapy, a person does not experience a noticeable decrease of symptoms, an increase in tolerance to the offending allergen and a reduction in medication usage. Treatment will not be reimbursed after a 2-year period when there is no apparent clinical benefit.

For those patients who have equivocal testing on IgE specific antibodies but have a strong clinical suspicion of allergic rhinitis and have a positive reaction to nasal allergen challenge, SCIT may be considered on a case-by-case basis as reasonable and necessary.

Provider Requirements

Payment may be made for a reasonable supply of antigens that have been prepared for a particular patient when:

The antigens are prepared by a physician who is a Doctor of Medicine or Osteopathy; and

The physician who prepared the antigens has examined the patient and has determined a plan of treatment and a dosage regimen; and because the major risk of AIT is anaphylaxis, SCIT should, therefore, be administered under the supervision of an appropriately trained physician who can recognize early symptoms and signs of anaphylaxis and administer emergency medications where necessary. In addition, SCIT should be administered only in facilities equipped to treat anaphylaxis.

It may be appropriate to permit patient self-administration at home for the patient with a history of life-threatening anaphylaxis who cannot receive immunotherapy in a health care facility. This requires very careful consideration of potential benefits and risks and should be made on an individual patient basis with appropriate informed consent.

Title XVIII of the Social Security Act, Sec.1862 (a)(1)(A) allows coverage and payment for only those items or services that are considered to be medically reasonable and necessary for the diagnosis or treatment of illness or injury or to improve the functioning of a malformed body member.

CMS Internet-Only Manual, Pub. 100-02, Medicare Benefit Policy Manual, Chapter 15 , Sec.50.2 Determining Self-Administration of Drug or Biological and Sec.50.4.4.1 Antigens

CMS Internet-Only Manual, Pub. 100-02, Medicare Benefit Policy Manual, Chapter 16, Sec.90 Routine Services and Appliances

CMS Internet-Only Manual, Pub. 100-03, Medicare National Coverage Determinations (NCD) Manual, Chapter 1, Part 2,

Sec.110.9 Antigens Prepared for Sublingual Administration and Sec.110.11 Food Allergy Testing and Treatment
CMS Internet-Only Manual, Pub. 100-04, Medicare Claims Processing Manual, Chapter 12, Sec.200 Allergy Testing and
Immunotherapy